Machine-Learning Assisted Materials Discovery of a Blue Emitter for a More Efficient and Durable OLED Device

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Abstract

While a more efficient and stable emitter with a deep blue color is essential for a better performance of organic light-emitting diodes (OLED) devices, obtaining such materials is extremely challenging. In the current work, we have developed a tool to assist the discovery of blue OLED materials by applying a v-HTS of materials based on the properties predicted using ML and a ML-predictabiltiy scale. In this study, the materials of interest include both Pt (II) phosphor complexes [1-3] and thermally activated delayed fluorescence materials for blue dopants [4-6]. We have also developed a user-friendly version that includes most of the algorithms used in this study, which is available on an internal web-based simulation platform.

Author Keywords

Machine learning; OLED; emitting materials; phosphors; thermally activated delayed fluorescence; and TADF.

1. Introduction

In the organic light-emitting diodes (OLED) industry, achieving efficient and stable blue dopant materials is one of the most important factors to manufacture more efficient and durable OLED devices, but obtaining such materials requires tremendous time as well as extensive monetary and human resources. Such new materials discovery involves three steps in cycle:

(1) Selection and design of candidate materials that are expected to exhibit desirable electronic properties,

(2) Computational validation and screening based on density functional theory (DFT) calculations, and

(3) Synthesis of materials and measurement of device properties.

These steps require solid understanding of molecular structures and properties to design new candidate materials, abundant computational resources to verify the target properties of various materials using ab-initio density functional theory (DFT) calculations, and sufficient experience and skill to successfully synthesize the designed molecules. Since both the second and third steps are expensive to perform for numerous candidate materials, the first step, a wise selection and design of candidate materials, is especially important. For example, the total number of candidate molecules can easily be over 1.5 million for one structure type of interest within the family of cyclometalated Pt (II) phosphor or thermally activated delayed fluorescence (TADF) materials. Therefore, in this study, we have developed a software tool that can assist the selection and design of candidate molecules to expedite this process by using machine learning (ML) of the molecular structure and electronic property relationships.

2. Method

Figure 1 describes the workflow of a ML-assisted virtual high-throughput screening (v-HTS) of materials that are applied in the current work. As a first step of this study, we have generated a full chemical materials space of interest suggested by organic chemists who are experts in the synthesis of organic compounds. Then, the candidate molecules are screened based on a ML-predictability scale followed by target properties predicted by ML property prediction models. The molecules with desirable ML-predicted properties are then examined using DFT calculations for further verification before being recommended further for synthesis.

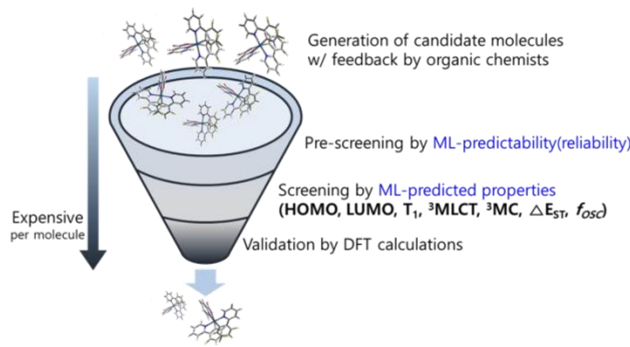
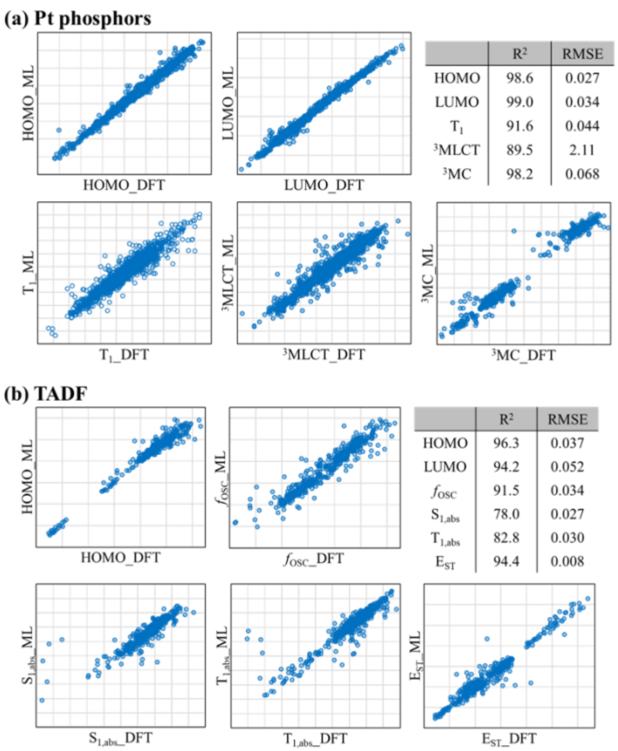


Figure 1. Overall workflow of a ML-assisted virtual high-throughput screening of materials, applied in the current work.

To build the ML property-prediction models, we selected 1,000 and 600 molecules among the candidate molecules for Pt phosphors and TADF materials, respectively, and obtained a database of properties by automating DFT calculations. Then, we built the ML models by calculating the numerical descriptors of molecules, preprocessing data, and employing cross validations and ensembles of the several best-performing ML models. The ML models include linear, non-linear and artificial neural network models using the Scikit-learn package [7] and Keras [8]. Hyper-parameters of the models are optimized by random search and Bayesian optimization. The ensemble models consist of the best three models that yield the lowest average value of root-mean-square errors (RMSE) across the 10-fold cross validations. For the predicted target electronic properties, we have examined the energy levels of the highest occupied molecule orbital (HOMO), the lowest un-occupied molecular orbital (LUMO), a triplet metal-ligand charge transfer state (^{3}MLCT) and a metal-centered triplet excited state (^{3}MC) for phosphors; and the energy levels of HOMO, LUMO, a singlet excited state ($S_{1,\mathrm{abs}}$), a triplet excited state ($T_{1,abs}$) and the splitting (ΔE_{ST}) between $S_{1,abs}$ and $T_{1,abs}$, and a oscillation strength (f_{OSC}) for TADF materials.

3. Results & Discussion

Figure 2 shows the cross-validation performance of the ML models. We have achieved the 10-fold average RMSEs of 0.025eV, 0.028eV, 0.043eV, 2.18% and 0.036eV for HOMO, LUMO, T_1, ^{3}MLCT and ^{3}MC, respectively, for Pt phosphors; and 0.037eV, 0.052eV, 0.034, 0.027eV, 0.030eV and 0.008eV for HOMO, LUMO, f_{OSC}, $S_{1,abs}$, $T_{1,abs}$ and ΔE_{ST}, respectively, for TADF materials. Thus, the cross-validation performance shows the ML models can accurately predict the properties of other molecules assuming the molecule of interest belongs to a group of molecules that are similar to what we trained. However, one question still remains: how reliably can we infer the properties of all of molecules across the whole materials space of interest by using the ML models built with only a small portion of materials? To answer the question, we have introduced the ML-predictability scale.



	R^2	RMSE
HOMO	98.6	0.027
LUMO	99.0	0.034
T_1	91.6	0.044
^{3}MLCT	89.5	2.11
^{3}MC	98.2	0.068

	R^2	RMSE
HOMO	96.3	0.037
LUMO	94.2	0.052
f_{osc}	91.5	0.034
$S_{1,abs}$	78.0	0.027
$T_{1,abs}$	82.8	0.030
E_{ST}	94.4	0.008

Figure 2. 10-fold cross validation performance of the machine-learning models for the DFT property prediction (a) for Pt phosphors and (b) for TADF materials.

We define the ML-predictability scale as a function comparing the compositions and structural information of candidate molecules relative to those of the molecules used to train ML models. The ML-predictability can be calculated relatively quickly within a half second per molecule, and a higher ML-predictability exhibits a higher probability of more accurate ML-predicted properties, as shown in Figure 3. Therefore, we have used the ML-predictability to quantitatively determine whether the ML models can infer the properties of a molecule reliably or not. Based on this data, we pre-screened candidate molecules before property predictions using ML.

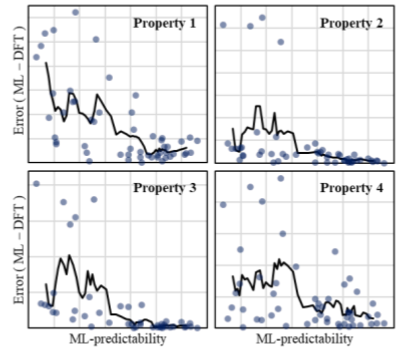


Figure 3. ML-prediction error with respect to DFT calculation values as a function of ML-predictability for different properties. Grey dots indicate individual molecules; and black solid lines indicate the averaged prediction error of six molecules that have consecutive ML-predictability values.

The choice of the cutoff value for the ML-predictability depends on the trade-off relations between the number of candidates to pre-screen and the prediction error after pre-screening. If we want to explore as many molecules as possible, we will need to sacrifice a high accuracy of ML property prediction. Figure 4 shows a two-dimensional map of the overall chemical space of interest mapped by t-distributed stochastic neighbor embedding (t-SNE) of molecular fingerprints. If we set the intermediate level of predictability as a cutoff criterion to pre-screen, we will examine a half (52%) of the total 1.5million candidate molecules to further screen based on the ML-predicted target properties, as shown in Figure 4. With employing pre-screening by our ML-predictability scale & screening of target properties based on the ML-predicted properties, we have narrowed down the candidate molecules and have further validated with actual DFT properties.

In Figure 5, the scatter plots and distribution plots show how the distribution of the DFT properties changes for the candidate molecules of chemists' interest with a certain ligands selection versus the candidate molecules predicted to be in the target range of our interest. The target-in rate of DFT properties (both T_1 and ^{3}MLCT) is doubled by implementing the v-HTS based on the ML-predicted properties from 22% to 48%. The target-in rate of DFT properties can further increases to 66%, if we employ a higher criterion for the ML-predictability scale to pre-determine the reliability of ML models for each candidate molecule, as shown in Figure 5.

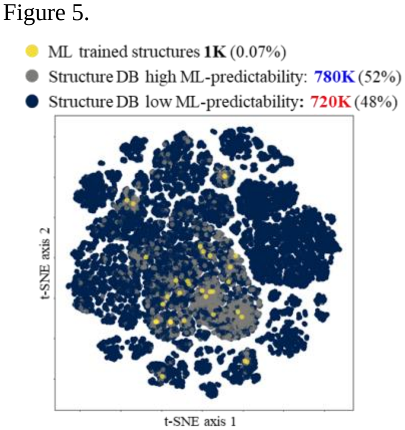


Figure 4. An overall map of candidate materials space

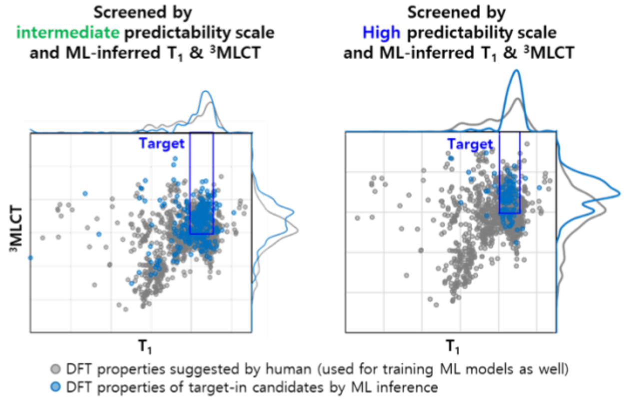


Figure 5. Scatter and distribution plots of T_1 and ^{3}MLCT of Pt phosphors, validated by DFT calculations. Grey dots and lines are for the candidate molecules of researchers' interest with a certain ligand selections; and blue dots and lines are for the candidate molecules predicted to be in the target range of our interest by the ML models.

4. Conclusion

We have successfully implemented a v-HTS of candidate materials based on the ML-predicted properties and the ML-predictability scale for both Pt phosphors and TADF materials for blue dopants. Further works involve the iteration of building a database of target properties and property-prediction models for the rest of candidate structures, thereby expanding the coverage of ML-prediction, as shown in Figure 6.

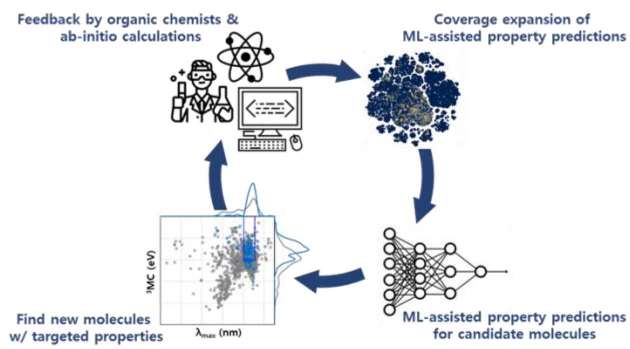


Figure 6. Overall process of materials discovery have implemented in the current work, and the iteration of the process to be implemented in the future.

5. References

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